University of Salzburg Lecturer: Roland Kwitt

Machine Learning (911.236), **Statistical Learning Theory** (536.505), **Advanced Machine Learning** (911.936)

Exercise sheet **B**

Exercise 1. 5 P.

Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0,1\}$. Assume that $\mathcal{H}$ is agnostic PAC learnable, i.e., there exists a learning algorithm A and a sample complexity function

$$m_{\mathcal{H}}^{\text{agn}} : (0,1)^2 \to \mathbb{N}$$

such that for every $\epsilon, \delta \in (0,1)$, for every probability distribution $\widetilde{D}$ on $\mathcal{X} \times \{0,1\}$, and for every sample size $m \geq m_{\mathcal{H}}^{\text{agn}}(\epsilon, \delta)$, the following holds: if

$$S = ((x_1, y_1), \ldots, (x_m, y_m)) \sim \widetilde{D}^m,$$

then, with probability at least $1 - \delta$ (over the choice of S),

$$L_{\widetilde{D}}(A(S)) \leq \inf_{h \in \mathcal{H}} L_{\widetilde{D}}(h) + \epsilon,$$

where

$$L_{\widetilde{D}}(h) = \mathbb{P}_{(x,y) \sim \widetilde{D}}[h(x) \neq y].$$

Show that $\mathcal{H}$ is PAC learnable in the realizable setting.

More precisely, prove that there exists a learning algorithm A' and a sample complexity function $m_{\mathcal{H}}^{\text{real}}$ such that for every $\epsilon, \delta \in (0,1)$, every probability distribution D on $\mathcal{X}$, every measurable target function (true labeling function)

$$f : \mathcal{X} \longrightarrow \{0,1\}$$

and every sample size

$$m \geq m_{\mathcal{H}}^{\text{real}}(\epsilon, \delta),$$

the following holds: if

$$S = ((x_1, f(x_1)), \ldots, (x_m, f(x_m)))$$

with $\forall i : x_i \sim D$, then, with probability at least $1 - \delta$ (over the choice of S),

$$\mathbb{P}_{x \sim D}[A'(S)(x) \neq f(x)] \leq \epsilon.$$

Important. Your proof should handle the following two cases (in a formally precise manner):

1. $\mathcal{X}$ is finite or countable,
2. $\mathcal{X}$ is uncountable (for example $\mathcal{X} = \mathbb{R}^n$).

In particular, if $\mathcal{X}$ is finite or countable, it is acceptable to define the induced joint distribution on $\mathcal{X} \times \{0,1\}$ pointwise. If $\mathcal{X}$ is uncountable, you should define that distribution measure-theoretically, for example via a pushforward measure or via its action on measurable sets.

Total #points: 5 P.